

A Packing of 12 Four-Spaces in $\mathbb{R}^{10}$

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October 9, 2026

Abstract

We give 12 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.617891. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.599696, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 12 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.617891 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 12$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.599696, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 12/11 = 2.618182$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N12.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 62 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
118995059624615352450289830270623915611076051186594490967361407678383878
334105296122904909578184865765599809159737598067198375107560451704058415
366153900141193167771792980893163981669375461875
```

denominator:

```
454545454545426049125102164947632866746621801756969507351470060823451046
451230653261164594588030141093319307819812193595732150494259228917322940
07842100220859645238650010530786081430617906884
```

B Coordinates

12 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N12.txt`.

```
-142273599615 21349369736 -397941664287 -90712199734270976349997 -225736054199 -159478161481 -70957848455460602566 -271182427241 -390128964708 -245384985000
310217070592 174802877152 -203297180604 -1570939525325197887469 -89893156346 48720151312 -843317662753886392303 -19537004346 -716932690543 77855458159
122669049867 -59001831171 -34548935646 103114386374268718271404 -3345863234 671529340244 17644670736141792702319 -478429985813 448434231209 -41246768501
-409060552573 -740755838622 -148556178873 21863681423371936768 -601487105099 232430294846 -2498852281222661640090 -338994869746 169017474373 -316686781617
-331466016073 100104531615 -12560047970 -57785981733111416471 115687421492 202207176865 -7460007240401598443574 293723036819 -9457743297 -356546497187
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